

Multijet balance: finding the data/MC $\langle x_j \rangle$ bias

Skim, sample stitching and JER smearing of the recoil jets

Samuel F. Liechty

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The observable and the problem

- Multijet balance $x_j = p_{T,\text{lead}}/|\vec{p}_{T,2} + \vec{p}_{T,3}|$, in bins of leading-jet p_T , seven jet radii.
- Selection: leading ≥ 20 GeV, jets 2 and 3 in $[7, 30)$ GeV, $|\eta| < 1.1 - R$, $\Delta\phi_{12} > 3\pi/4$, $\Delta\phi_{13} > \pi/2$.
- MC jets get an extra JER smearing, so that MC resolution matches data.
- With the original trees, data/MC of $\langle x_j \rangle$ was far from 1 ($R = 0.4$):

leading p_T (GeV)	20–25	25–30	30–35	35–40	40–50
data/MC	1.077	1.067	1.063	1.037	1.034

- We expect this ratio to be within about 1% of unity.
- The trigger requirement and a data JES of 0.93 were both tried and did not fix it.

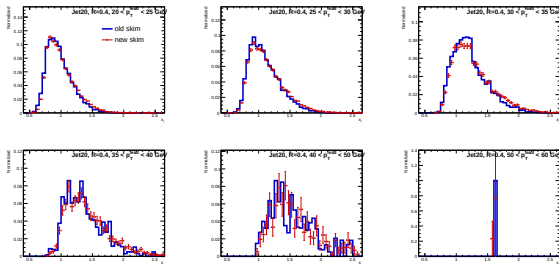
Finding 1: the skim cut MC on a different p_T than the analysis

- The treemaker skimmed events on the unsmeared calibrated p_T (leading ≥ 20 , jets 2 and 3 ≥ 7 GeV).
- It also kept only the top three jets by calibrated p_T .
- The analysis cuts MC on the smeared p_T .
- Events that pass only after smearing were never written.
- These come mostly from the lower $\hat{p}_T$ samples and have high x_j .
- Cross sections and stitching thresholds were checked and are correct.

Fix in DijetTreeMaker:

- MC: keep the event if any of the seven p_T flavors passes (calib + six JER smears).
- Keep the union of each flavor's top three jets.
- Store jets if any flavor ≥ 5 GeV.
- Data: skim thresholds $\times 0.9$, for in-situ JES scans down to $p_a = 0.9$.

The new skim recovers 3–4 $\times$ more selected MC events

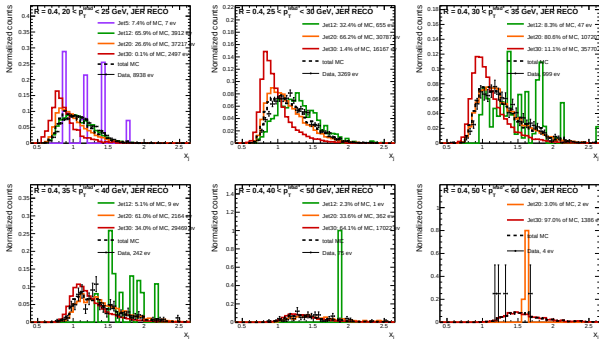


Jet20 sample, $R = 0.4$, same analysis code:

p_T bin	events	$\langle x_j \rangle$
20–25	$\times 4.2$	$0.975 \rightarrow 1.001$
25–30	$\times 3.6$	$1.097 \rightarrow 1.130$
30–35	$\times 3.3$	$1.219 \rightarrow 1.267$
35–40	$\times 3.1$	$1.376 \rightarrow 1.421$

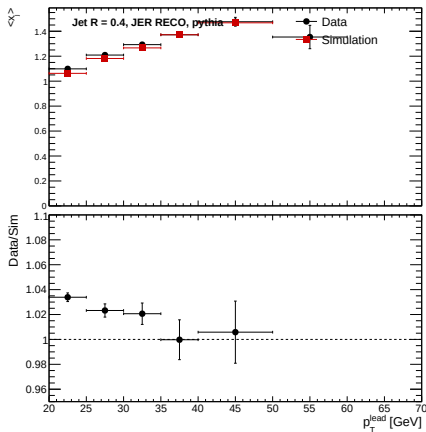
- About two thirds of the selected MC events were missing.
- The recovered events sit in the high- x_j tail.

Finding 2: Jet5 adds noise, so stitch from Jet12



- At fixed leading p_T , lower $\hat{p}_T$ samples have higher x_j .
- The MC mean is therefore sensitive to the sample mix.
- Jet5 carries 9.6% of the 20–25 GeV weight from 9 events.
- In 25–30 GeV one Jet5 event carries 6.1%.
- Jet8 has no MDC2 thresholds to stitch with Jet5.
- Jet5 and Jet8 are removed, and Jet12 starts at truth $p_T = 0$.

After the new skim: a 2–3% excess remains at low p_T



New MC trees, Jet12/20/30, R = 0.4, reco-anchored smearing:

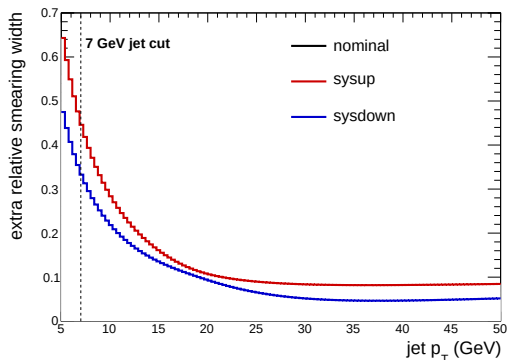
	20–25	25–30	30–35	35–40	40–50
data/MC	1.034	1.023	1.021	1.000	1.006

Data vs weighted MC, 20–25 GeV bin:

quantity	data/MC
$\langle p_{T,\text{lead}} \rangle$	1.000
$\langle p_{T,2} \rangle$	0.935
$\langle p_{T,3} \rangle$	0.956

- The reweighting closes; the MC recoil jets are too hard.

Finding 3: the extra smearing is very wide for soft jets



- Template
h_jer_smear_r04_pileup_EMfracJES,
used for every radius.
- Extra width: 39% at 7 GeV, 25% at 10 GeV, 10% at 20 GeV.
- It is added on top of the MC's own resolution.
- Soft jets on a steep spectrum are pushed over the 7 GeV cut.

Test with no extra MC smearing ($R = 0.4$):

	20–25	25–30	30–35	35–40
smear	1.034	1.023	1.021	1.000
no smear	0.991	0.997	1.000	0.991

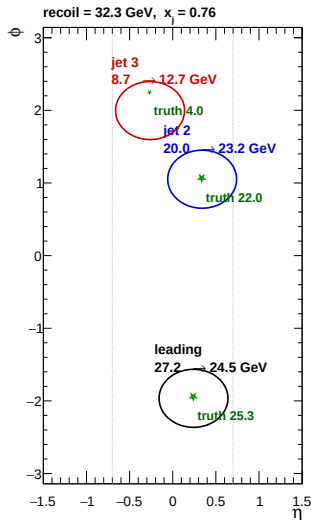
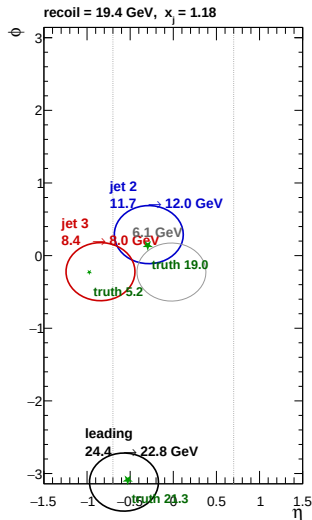
The third jet usually has no truth jet above 7 GeV

Jet20, events passing the analysis 3-jet selection (smeared jets): fraction with ≤ 2 truth jets.

R	all stored truth jets (≥ 3 GeV)	truth ≥ 7 GeV	truth ≥ 7 GeV in acceptance
0.2	13.5%	80.1%	82.5%
0.3	16.3%	76.4%	81.7%
0.4	16.6%	71.0%	81.4%
0.5	16.2%	66.3%	83.4%
0.6	17.5%	65.8%	87.6%

- In most selected MC events, jet 3 is a soft jet fluctuated over 7 GeV.
- About 1 in 6 has no third truth jet at all, so it is a fake or a split jet.
- This is why the result depends so strongly on the low- p_T smearing width.

Event displays: the events we are correcting (1/3)

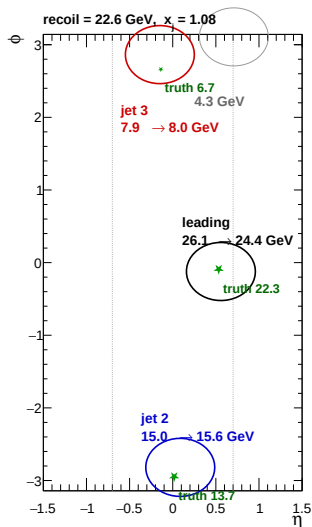
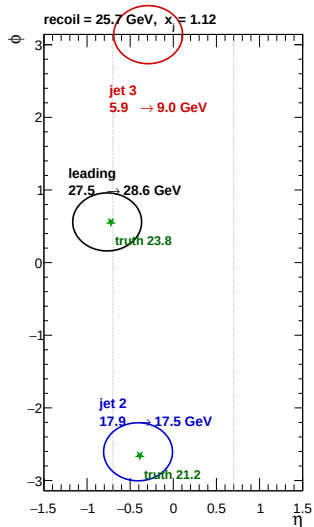


Pythia8 Jet20, $R = 0.4$: selected events with three reco jets but only two truth jets ≥ 7 GeV.

Reco jets show calibrated $\rightarrow$ smeared p_T ; green stars are truth jets; dotted lines mark the acceptance.

- Left: jet 3 (8.4 GeV) sits on a 5.2 GeV truth jet.
- Right: jet 3 is smeared from 8.7 to 12.7 GeV on a 4.0 GeV truth jet.

Event displays: the events we are correcting (2/3)

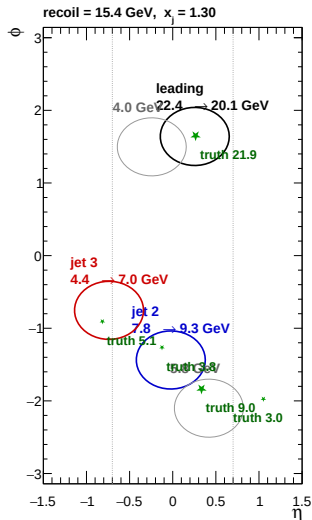
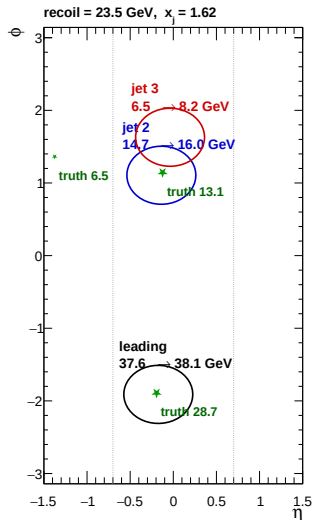


Pythia8 Jet20, $R = 0.4$: selected events with three reco jets but only two truth jets ≥ 7 GeV.

Reco jets show calibrated $\rightarrow$ smeared p_T ; green stars are truth jets; dotted lines mark the acceptance.

- Left: jet 3 has no truth jet and is smeared from 5.9 to 9.0 GeV.
- Right: jet 3 (7.9 GeV) sits on a 6.7 GeV truth jet, just above the cut.

Event displays: the events we are correcting (3/3)



Pythia8 Jet20, $R = 0.4$: selected events with three reco jets but only two truth jets ≥ 7 GeV.

Reco jets show calibrated $\rightarrow$ smeared p_T ; green stars are truth jets; dotted lines mark the acceptance.

- Left: jet 3 overlaps jet 2, so it is a split piece of the 13.1 GeV truth jet.
- Right: smearing pushes two soft jets over 7 GeV.
- The 9.0 GeV truth jet's reco partner (5.8 GeV) is not selected.

How often this happens in the selected MC

Fraction of selected events with exactly two truth jets ≥ 7 GeV (current selection, stitched Jet12/20/30).

R	Jet12	Jet20	Jet30	cross-section weighted MC
0.2	65.2%	73.2%	62.8%	68.3%
0.3	73.1%	72.3%	54.3%	72.3%
0.4	77.3%	68.7%	45.8%	73.3%
0.5	74.0%	54.1%	28.1%	72.1%
0.6	74.8%	52.8%	23.7%	73.7%
0.7	73.8%	46.5%	18.2%	73.3%
0.8	70.1%	50.5%	50.0%	70.0%

- About 70% of the weighted MC sample, at every radius.
- The lower $\hat{p}_T$ samples, which dominate the low- p_T bins, have the most.
- These are the events the new recoil smearing is applied to.

Proposed correction: smear the recoil, not the soft jets

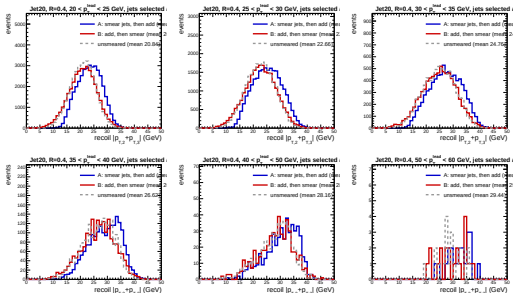
- $\text{Recoil} = |\vec{p}_{T,2} + \vec{p}_{T,3}|$.
- Current: smear jets 2 and 3 individually, then add them.
- Proposed, in events with exactly two truth jets ≥ 7 GeV: add jets 2 and 3 unsmeared, then smear the sum once.
- The width comes from the same template, evaluated at the p_T of the truth jet matched to the recoil.
- The recoil truth jet is the one not matched (in ΔR) to the leading reco jet.

Jet20, $R = 0.4$, 20–25 GeV, two-truth-jet events, mean x_j :

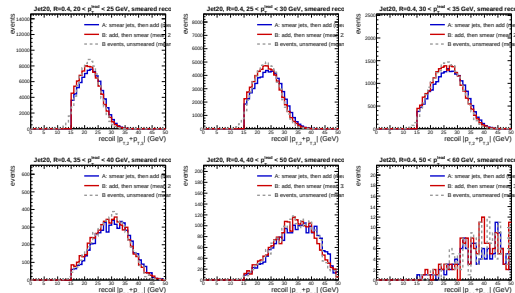
comparison	unsmeared	A: smear, then add	B: add, then smear
same jets, cuts before smearing	1.063	1.084	1.076
jets selected after smearing	1.137	1.002	1.143
each cut on its own smeared recoil ≥ 15 GeV	–	1.044	1.067

Recoil spectra: smear-then-add vs add-then-smear

Jets selected after smearing



Each method cut on its own smeared recoil ≥ 15 GeV



- Two individual smears at low- p_T widths give a broader, harder recoil.
- Near a cut on a falling spectrum, that means more upward migration.

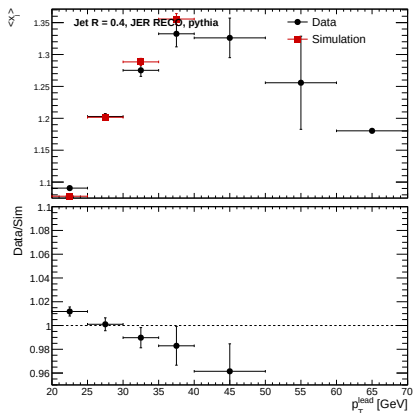
A recoil-only cut needs to stay inside both skims

- Dropping the per-jet 7 GeV cuts gave data/MC ≈ 0.96 – 0.97 .
- Jet 3 below 7 GeV: 70% of selected MC events, but only 34–40% in data.
- The data skim requires three calib jets ≥ 7 GeV; the MC skim accepts any flavor.
- So the event sample was defined by the two different skims.

Adopted selection (option 1):

- Leading ≥ 20 GeV, jets 2 and 3 ≥ 7 GeV each, recoil ≥ 14 GeV.
- The 7 GeV cuts use each event's values before any recoil smearing.
- Two-truth-jet MC events use the recoil smearing; all others keep individual smearing.

Result: $R = 0.2\text{--}0.5$ within about 1% below 35 GeV



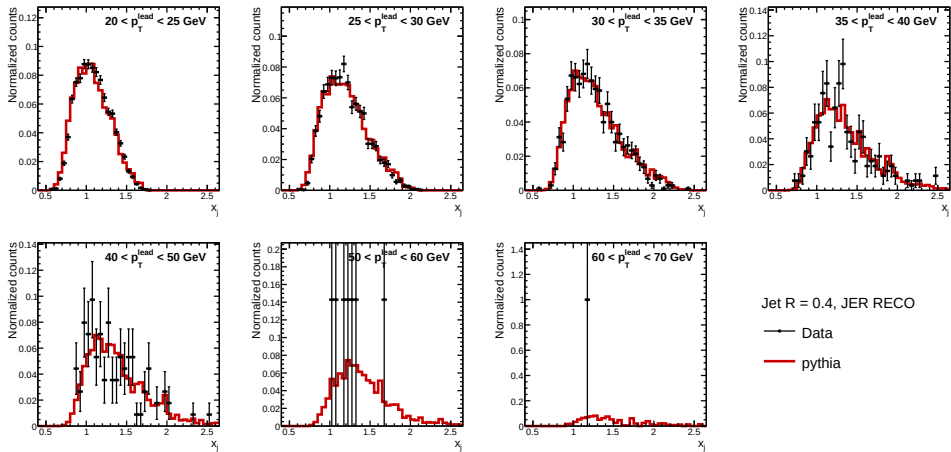
$R = 0.4$, data/MC of $\langle x_j \rangle$:

	20–25	25–30	30–35	35–40	40–50
original	1.077	1.067	1.063	1.037	1.034
reco smear	1.012	1.001	0.990	0.983	0.961
truth smear	1.007	0.997	0.993	0.983	0.958

20–25 GeV bin, all radii (reco smear):

0.2	0.3	0.4	0.5	0.6	0.7	0.8
0.998	1.002	1.012	1.001	1.020	1.026	1.041

Result: x_j distributions, $R = 0.4$



- Data and MC shapes agree in every bin up to 40 GeV.
- The JER high/low variations bracket the nominal within about $\pm 0.5\%$.

Summary and next steps

- The skim cut MC on calibrated p_T , while the analysis cut on smeared p_T ; this is now fixed in the treemaker.
- Jet5 and Jet8 are removed, and Jet12 starts at truth $p_T = 0$.
- The large low- p_T smearing width drove the remaining 2–3% excess through soft jets crossing 7 GeV.
- Smearing the summed recoil in two-truth-jet events, with a 14 GeV recoil cut, brings $R = 0.2$ – 0.5 to within about 1%.

Open items

- Provenance of the JER template: extra width or total data resolution?
- The same template feeds the γ +jet unfolding, so check its impact there.
- The ratio falls with leading p_T (0.96 at 40–50 GeV, $R = 0.4$); watch it with more statistics.
- $R \geq 0.6$ is still 2–4% high at low p_T , and its template is $R = 0.4$.
- Data trees: rerun with the finished reprocessing; herwig still to be regenerated.